

EffiSurg: Autonomous Microsurgical Robotic Manipulation via Efficient Flow Matching

Author (s)

Abstract—Generative action policies based on diffusion or flow matching offer a promising approach to autonomous microsurgical manipulation, but their iterative generation incurs substantial inference overhead. EffiSurg is an efficient flow-matching policy that decouples multimodal grounding from iterative action refinement. It extracts action-specific conditioning features from multimodal representations once per predicted action chunk. Conditioned on these features, a lightweight flow expert iteratively refines actions without repeated interaction with the full multimodal representations. EffiSurg is evaluated on a bimanual microsurgical experimental platform across four tasks involving needle handoff and knot tying in phantom and ex vivo settings. Compared with a coupled transformer-based flow-matching baseline, EffiSurg reduces end-to-end policy inference latency by 67% to 49 ms while increasing the average task success rate by 16.3 percentage points to 96.3%. These results demonstrate the potential of EffiSurg to provide computationally efficient action generation for autonomous surgical systems, supporting timely integration of perceptual feedback into closed-loop microsurgical manipulation. Project website: <https://anonymous.4open.science/w/ICRA2027-EffiSurg/>.

I. INTRODUCTION

Autonomous microsurgical manipulation requires precise coordination of surgical instruments when handling small needles and deformable sutures in a confined workspace. Such manipulation relies on visual and proprioceptive feedback to guide instrument interactions and adjust actions as the surgical scene changes. Policy inference introduces computational latency between receiving observations and producing action predictions. Reducing this latency while maintaining effective action generation is therefore an important challenge for autonomous microsurgical manipulation.

Imitation learning has become a practical approach for acquiring precise visuomotor policies from expert demonstrations [1], [2]. Action-chunking methods jointly predict short sequences of future actions, capturing temporal structure for fine-grained bimanual manipulation [3], [4]. Generative policies further model conditional distributions over action chunks, allowing multiple valid behaviors to be represented [5], [6]. Diffusion- and flow-based policies generate these chunks through iterative refinement, but repeated denoising or vector-field evaluations introduce substantial inference overhead [5], [7], [8].

In architectures that couple multimodal grounding with iterative action refinement, each generation step involves interactions between evolving action representations and high-dimensional multimodal representations. During the generation of a single action chunk, the conditioning observation remains fixed, while the action representations and generation time evolve. This raises the question of

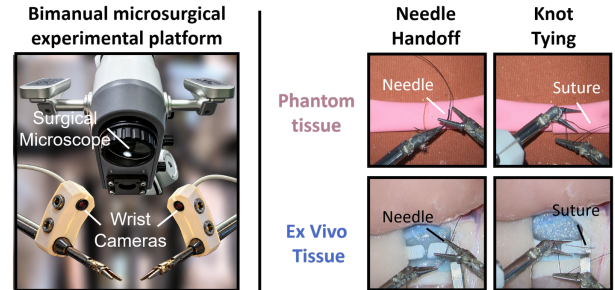


Fig. 1. Overview of the bimanual microsurgical experimental platform and evaluation tasks. (Left) The bimanual microsurgical experimental platform is equipped with a surgical microscope and two wrist cameras. (Right) EffiSurg learns needle handoff (left column) and knot tying (right column) in phantom (top row) and ex vivo (bottom row) settings, yielding four evaluation tasks.

whether action-relevant information can be extracted before refinement and reused throughout generation. The challenge is to preserve sufficient multimodal guidance for each future action without repeatedly interacting with the full multimodal representations. Rather than reducing the number of generation steps, this work addresses that challenge by reorganizing the computation within the action-generation process.

To this end, an efficient flow-matching framework, **EffiSurg**, is proposed that decouples multimodal grounding from iterative action refinement. A single shared observation feature may provide insufficiently differentiated guidance across future-action positions. EffiSurg therefore uses learned action queries to extract action-specific conditioning features once per predicted chunk, with one feature corresponding to each future action. A lightweight flow expert then iteratively predicts the vector field under these fixed conditions. Action-specific features and flow time jointly modulate shared MLP blocks, while an intermediate self-attention layer enables information exchange across the action chunk. This design moves multimodal grounding outside the iterative refinement loop and uses a lightweight action generator to reduce per-step computational cost while preserving multi-step generation.

EffiSurg is evaluated on the bimanual microsurgical experimental platform shown in Fig. 1 across four needle-handoff and knot-tying tasks in phantom and ex vivo settings, using approximately 600 expert demonstrations. Comparisons include Diffusion Policy, Flow Matching Policy, and a coupled Transformer Flow Policy with matched visual encoder and multimodal transformer configurations. Relative to the coupled baseline, EffiSurg reduces end-to-end policy

inference latency by 67% to 49 ms per action chunk while increasing average task success rate by 16.3 percentage points to 96.3%.

The main contributions are summarized as follows:

- An efficient flow-matching framework is proposed for autonomous microsurgical manipulation. By extracting action-specific conditioning features once per action chunk, the framework decouples multimodal grounding from iterative action refinement, reducing inference overhead while preserving multi-step action generation.
- A lightweight flow expert is developed with condition-modulated MLP blocks for efficient action refinement and an intermediate self-attention layer for modeling dependencies across future actions, supporting coordinated action generation under action-specific conditioning.
- A bimanual microsurgical experimental platform is set up, and approximately 600 synchronized multimodal demonstrations are collected for needle handoff and knot tying in phantom and ex vivo settings. Evaluations across four tasks examine policy efficiency and effectiveness through baseline comparisons and architectural ablations.

II. RELATED WORK

A. Generative Imitation Learning

Imitation learning enables robotic policies to acquire manipulation skills directly from expert demonstrations [1], [2], [9]. Action-chunking methods such as ACT model temporally extended actions for fine-grained bimanual manipulation [3]. Diffusion Policy performs iterative action denoising [5], [10], while RDT-1B scales diffusion transformers to bimanual manipulation [6]. Flow-matching policies learn continuous vector fields for iterative action generation, with Affordance-Based Robot Manipulation with Flow Matching providing an early formulation for robot manipulation [7], [8]. However, these iterative generative policies incur substantial inference costs. To address this challenge, recent studies have explored reducing the number of sampling steps or the computational cost per step [11]–[14]. Yet, these acceleration methods have primarily been evaluated in tabletop manipulation settings, while microsurgical manipulation poses additional challenges in precision, data availability, and sensing.

B. Autonomous Surgical Manipulation

Imitation learning has been applied to surgical tasks under explicit motion constraints [15], while the SRT series extended end-to-end transformer policies to complex procedures such as tissue manipulation, needle handling, and knot tying [16], [17]. Subsequent studies have further improved robustness, precision, data efficiency, and spatial understanding for diverse surgical tasks [18]–[23]. However, existing studies primarily focus on task performance, robustness, generalization, or long-horizon autonomy, with comparatively limited attention to policy inference efficiency.

III. PRELIMINARIES

This section reviews the standard formulation of conditional flow-matching action policies and establishes the notation used in the remainder of the paper. Given the current multimodal observation, such a policy generates a short-horizon action chunk that can be executed by the robot in closed loop. At control step t , the observation is denoted as

$$o_t = \{I_t^{1:V}, \mathbf{s}_t\}, \quad (1)$$

where $I_t^{1:V} = \{I_t^1, \dots, I_t^V\}$ represents the observations from V visual views, and $\mathbf{s}_t$ denotes the robot proprioceptive state. The policy predicts a future action chunk of horizon H ,

$$\mathbf{A}_t = [\mathbf{a}_t^1, \mathbf{a}_t^2, \dots, \mathbf{a}_t^H]^\top \in \mathbb{R}^{H \times d_a}, \quad (2)$$

where $\mathbf{a}_t^h \in \mathbb{R}^{d_a}$ denotes the action at the h -th future step.

Rather than directly regressing $\mathbf{A}_t$, a conditional flow-matching policy models the action distribution $p(\mathbf{A}_t | o_t)$ through a continuous transport process [8]. Let $\mathbf{A}_t^{(0)} \sim p_0$ denote an action chunk sampled from a simple prior distribution and let $\mathbf{A}_t^{(1)} \sim p_{\text{data}}(\mathbf{A}_t | o_t)$ denote the corresponding demonstration action chunk. The commonly used linear conditional probability path is

$$\mathbf{A}_{t,\tau} = (1 - \tau)\mathbf{A}_t^{(0)} + \tau\mathbf{A}_t^{(1)}, \quad \tau \in [0, 1]. \quad (3)$$

The sample-wise target vector field along this path is

$$\mathbf{u}_{t,\tau} = \frac{d\mathbf{A}_{t,\tau}}{d\tau} = \mathbf{A}_t^{(1)} - \mathbf{A}_t^{(0)}. \quad (4)$$

A conventional conditional flow-matching policy learns the vector field by minimizing

$$\mathcal{L}_{\text{FM}} = \mathbb{E} \left[\|v_\theta(\mathbf{A}_{t,\tau}, \tau, o_t) - \mathbf{u}_{t,\tau}\|_F^2 \right], \quad (5)$$

where the expectation is taken over training observations and demonstration actions, prior samples, and flow times $\tau \sim \mathcal{U}[0, 1]$. At inference time, the learned vector field defines the ordinary differential equation (ODE)

$$\frac{d\mathbf{A}_{t,\tau}}{d\tau} = v_\theta(\mathbf{A}_{t,\tau}, \tau, o_t), \quad (6)$$

where v_θ is a conditional vector field parameterized by the policy network.

During inference, the action chunk is obtained by numerically integrating the learned vector field from $\tau_0 = 0$ to $\tau_K = 1$, starting from $\mathbf{A}_t^{(0)} \sim p_0$. Using K uniformly spaced Euler integration steps, the update can be written as

$$\mathbf{A}_{t,\tau_{k+1}} = \mathbf{A}_{t,\tau_k} + \Delta\tau v_\theta(\mathbf{A}_{t,\tau_k}, \tau_k, o_t), \quad (7)$$

where $\Delta\tau = 1/K$ and $\tau_k = k/K$ for $k = 0, \dots, K$. Consequently, the vector field must be evaluated repeatedly during the generation of every action chunk.

In coupled multimodal flow policies, each vector-field evaluation involves interactions between the evolving action representations and high-dimensional multimodal representations. Consequently, the computational cost of iterative generation is determined not only by the number of integration steps, but also by the cost of multimodal interaction at each step. This repeated computation constitutes the primary bottleneck targeted in this work.

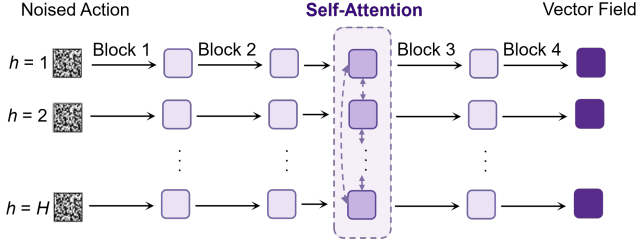


Fig. 3. Architecture of the action-space flow expert. Each evolving action feature is processed by four shared MLP blocks under its corresponding action-specific conditioning. A self-attention layer inserted between the second and third MLP blocks enables information exchange across future actions, after which the refined features are projected to the conditional vector field.

action, the evolving action is first projected into an expert feature of width d_e ,

$$\mathbf{z}_h^{(0)} = f_{\text{in}}(\mathbf{a}_{t,\tau}^h) \in \mathbb{R}^{d_e}. \quad (9)$$

For future action h at block ℓ , the corresponding conditioning feature $\mathbf{c}_{t,h}$ and the flow-time embedding $e(\tau)$ are jointly projected to produce feature-wise scale, shift, and residual-gating parameters,

$$[\alpha_h^{(\ell)}, \beta_h^{(\ell)}, \gamma_h^{(\ell)}] = g_\phi^{(\ell)}(\mathbf{c}_{t,h}, e(\tau)). \quad (10)$$

The position-wise update is then

$$\mathbf{z}_h^{(\ell)} = \mathbf{z}_h^{(\ell-1)} + \gamma_h^{(\ell)} \odot f_{\text{MLP}}\left(\alpha_h^{(\ell)} \odot \text{LN}(\mathbf{z}_h^{(\ell-1)}) + \beta_h^{(\ell)}\right), \quad (11)$$

for $\ell = 1, \dots, L$. Thus, the parameters within each block are shared across future-action positions, while each position is modulated by its own action-specific conditioning.

Although the MLP blocks efficiently process individual action features, future actions within a chunk are not independent. As illustrated in Fig. 3, a self-attention layer is inserted between the second and third MLP blocks to exchange information across the H action features. The third MLP block takes the self-attention-refined features as input. After the fourth block, an output projection maps each refined feature to its vector-field component, $\hat{\mathbf{u}}_{t,\tau}^h = f_{\text{out}}(\mathbf{z}_h^{(L)})$. In this way, repeated vector-field evaluations operate on evolving action representations and compact action-specific conditioning features without repeatedly revisiting the high-dimensional multimodal representations.

EffiSurg is trained with the conditional flow-matching objective

$$\mathcal{L}_{\text{FM}} = \mathbb{E} \left[\|\hat{\mathbf{u}}_{t,\tau} - \mathbf{u}_{t,\tau}\|_F^2 \right], \quad (12)$$

where the expectation is taken over $(o_t, \mathbf{A}_t^{(1)}) \sim p_{\text{data}}$, $\mathbf{A}_t^{(0)} \sim p_0$, and $\tau \sim \mathcal{U}[0, 1]$. As defined above, $\mathbf{A}_{t,\tau} = (1-\tau)\mathbf{A}_t^{(0)} + \tau\mathbf{A}_t^{(1)}$ and $\mathbf{u}_{t,\tau} = \mathbf{A}_t^{(1)} - \mathbf{A}_t^{(0)}$ denote the linear path and its sample-wise target vector field, respectively. During inference, the predicted vector field is integrated over flow time to progressively transform $\mathbf{A}_t^{(0)}$ into the final action chunk $\mathbf{A}_{t,1}$, while $\mathbf{C}_t$ remains fixed across all K vector-field evaluations.

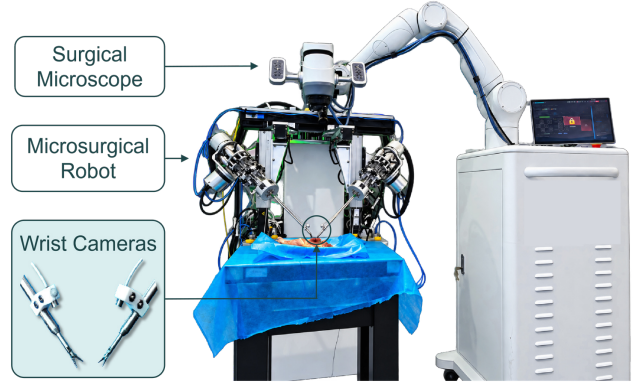


Fig. 4. Experimental platform. The custom-developed microsurgical robotic system comprises a surgical microscope and a microsurgical robot. One top view and two wrist views provide complementary visual observations of the surgical workspace.

V. EXPERIMENTS

The experiments address three questions:

- How much does EffiSurg reduce policy inference latency relative to the coupled transformer baseline?
- How does EffiSurg compare with baseline policies in task success across phantom and ex vivo settings?
- How do action-specific conditioning, intermediate self-attention, and the number of flow-integration steps affect performance and inference latency?

A. Experimental Setup

Experimental Platform. Experiments are conducted on the custom-developed bimanual microsurgical robotic platform shown in Fig. 4. Visual observations comprise one top view from the surgical microscope and two wrist views from wrist cameras, providing complementary global and local information about the surgical scene. Robot proprioceptive states are acquired synchronously with the visual observations and provided to the policy as additional input. The robotic system is equipped with an NVIDIA GeForce RTX 3080 GPU.

Tasks and Data Collection. EffiSurg is evaluated on two bimanual surgical skills—needle handoff and knot tying—in phantom and ex vivo settings, yielding four evaluation tasks, as shown in Fig. 5: *needle handoff on phantom tissue*, *knot tying on phantom tissue*, *needle handoff on ex vivo tissue*, and *knot tying on ex vivo tissue*. *Needle handoff* requires precise grasping, reorientation, and transfer of a surgical needle between two instruments, placing demands on fine bimanual coordination and reliable needle handling. *Knot tying* involves coordinated manipulation of a deformable suture whose configuration continuously changes during execution, requiring the policy to adapt its actions to evolving visual observations while maintaining temporal coordination between the two manipulators. The phantom setting provides a controlled environment for evaluating policy behavior, whereas the ex vivo setting introduces more realistic visual appearance, tissue interaction, and task variability. For each of the four evaluation tasks, approximately 150

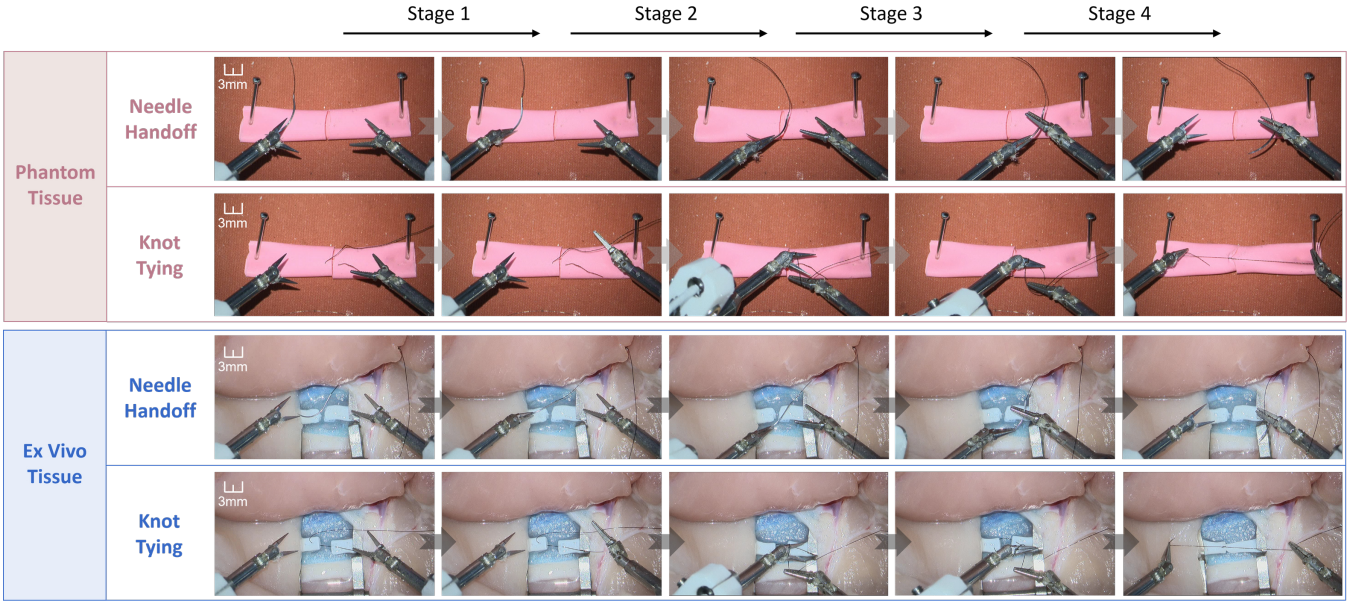


Fig. 5. Overview of the four surgical manipulation tasks used for evaluation: needle handoff on phantom tissue, knot tying on phantom tissue, needle handoff on ex vivo tissue, and knot tying on ex vivo tissue. Each task comprises four stages. The needle-handoff task includes *Needle Grasping*, *Needle Reorientation*, *Needle Transfer*, and *Secure Holding*; the knot-tying task includes *Suture Grasping*, *Loop Formation*, *Suture Regrasping*, and *Knot Tightening*. The instruments have a distal diameter of 2.8 mm, and 7-0 sutures are used in the phantom and ex vivo settings.

expert demonstrations are collected through teleoperation using omega.7 devices. During demonstration collection, the operator controls the two robotic manipulators through the devices, while multi-view RGB observations, robot proprioceptive states, and corresponding actions are synchronously recorded.

Baselines. EffiSurg is compared with *Diffusion Policy* [5] and *Flow Matching Policy* [7]. The CNN/U-Net-based visuomotor implementations used in the original works are adopted and adapted to the multi-view observations and robot action space in the experiments. To evaluate the proposed decoupling of multimodal grounding from iterative action refinement, a controlled baseline, termed *Transformer Flow Policy*, is further introduced. This baseline employs a transformer action expert that processes the evolving action chunk conditioned on the flow-time embedding. At each flow-integration step, the evolving action representations attend to the multimodal representations through cross-attention, and the expert predicts the conditional vector field, thereby coupling multimodal grounding with iterative action refinement. Transformer Flow Policy and EffiSurg use the same visual encoder and multimodal transformer architectures. The transformer action expert in the baseline matches the depth, hidden width, and number of attention heads of the action-specific conditioning transformer in EffiSurg.

Training and Inference Details. All methods use the same training demonstrations, with a separate policy trained for each of the four evaluation tasks. All policies are trained using the AdamW optimizer for 200 epochs per task with a global batch size of 128. The learning rate reaches a peak of 1×10^{-4} after 1,000 warm-up steps and subsequently follows a cosine decay schedule. The visual encoders in

both EffiSurg and Transformer Flow Policy remain frozen throughout training. At inference time, each policy predicts an action chunk of length $H = 16$. Only the first 10 actions are executed, after which a new action chunk is predicted using the latest multimodal observation. Flow-based policies use 10-step Euler integration, while Diffusion Policy uses 10-step DDIM sampling. All methods are evaluated on the same robotic platform and computing hardware.

Evaluation Metrics. Each policy is evaluated over 20 trials per task with varied needle positions and orientations and suture configurations. All methods follow the same initialization and reset procedures. Ex vivo tests use tissue specimens not used for demonstration collection, with each specimen used to evaluate all methods to reduce the influence of specimen variability. Each trial has a time limit of 30 s; trials that do not complete the task within this limit are terminated and counted as failures. Needle handoff is successful when the receiving instrument securely holds the needle after the delivering instrument releases it. Knot tying is successful when the prescribed knot is formed and tightened. Task success rate is the percentage of the 20 trials meeting the success criteria, and average success rate is the arithmetic mean across the four tasks. Cumulative stage completion rates are reported following Fig. 5, measuring the percentage of all 20 trials that complete the corresponding stage and all preceding stages. End-to-end policy inference latency is measured from receipt of the latest multimodal observation to generation of a complete action chunk.

B. Main Results

Performance on Phantom Tissue. As shown in Fig. 6 and Table I, EffiSurg successfully completes all 20 trials

TABLE I
TASK SUCCESS AND INFERENCE LATENCY ACROSS FOUR SURGICAL MANIPULATION TASKS.

Method	Iterative Action Refinement Steps	Phantom Tissue		Ex Vivo Tissue		Avg. Success Rate (%) $\uparrow$	Avg. Inference Latency (ms) $\downarrow$
		Needle Handoff	Knot Tying	Needle Handoff	Knot Tying		
Diffusion Policy [5]	10	11/20	12/20	9/20	9/20	51.3	109
Flow Matching Policy [7]	10	17/20	15/20	11/20	10/20	66.3	108
Transformer Flow Policy	10	18/20	17/20	15/20	14/20	80.0	147
EffiSurg	10	20/20	20/20	19/20	18/20	96.3	49

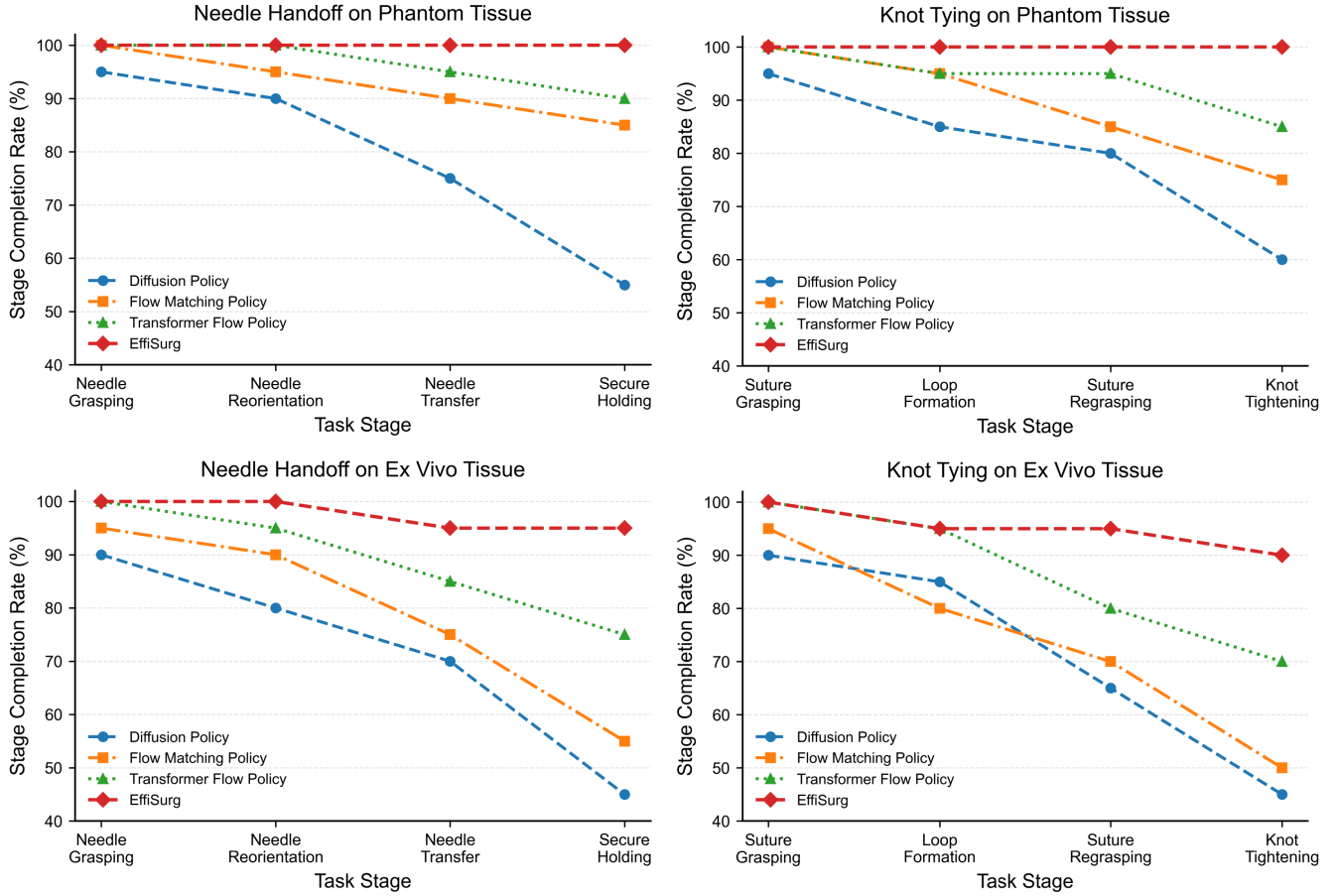


Fig. 6. Stage-wise completion rates for needle handoff (left) and knot tying (right) on phantom tissue (top) and ex vivo tissue (bottom). Each point represents the percentage of 20 trials that successfully complete all stages up to and including the indicated stage. The final-stage completion rate equals the overall task success rate reported in Table I.

for both needle handoff and knot tying on phantom tissue, achieving a 100% success rate on both tasks. In comparison, Diffusion Policy achieves 55% and 60%, Flow Matching Policy achieves 85% and 75%, and *Transformer Flow Policy* achieves 90% and 85% on needle handoff and knot tying, respectively. These results demonstrate reliable bimanual control for both surgical-needle transfer and deformable-suture manipulation, while the improvement over the coupled transformer baseline supports the effectiveness of the proposed decoupled design.

Performance on Ex Vivo Tissue. Compared with the phantom setting, the ex vivo setting introduces greater

variability in visual appearance, tissue configuration, and manipulation interactions, posing additional challenges to precise bimanual control. As shown in Fig. 6 and Table I, EffiSurg achieves success rates of 95% and 90% on needle handoff and knot tying, respectively, compared with 45% and 45% for Diffusion Policy, 55% and 50% for Flow Matching Policy, and 75% and 70% for *Transformer Flow Policy*. These results demonstrate that EffiSurg remains effective in a more realistic surgical manipulation environment and supports precise bimanual manipulation involving both rigid surgical instruments and deformable sutures.

Inference Efficiency. As shown in Table I, EffiSurg re-

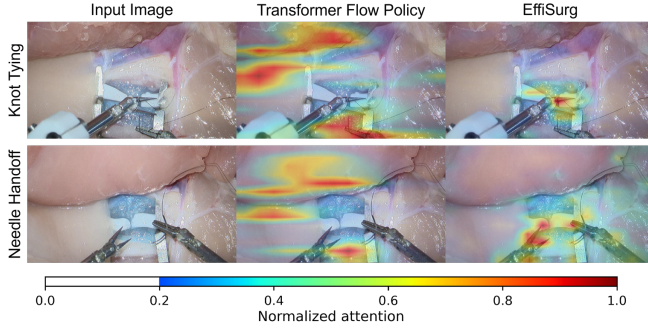


Fig. 7. Qualitative comparison of cross-attention to top-view tokens during surgical manipulation. Columns show the top-view input images (left), attention heatmaps of *Transformer Flow Policy* (middle), and attention heatmaps of *EffiSurg* (right). Heatmaps are overlaid on the same top-view images, with warmer colors indicating higher relative attention within the top view.

quires only 49 ms per action chunk, substantially lower than *Diffusion Policy* (109 ms), *Flow Matching Policy* (108 ms), and *Transformer Flow Policy* (147 ms). In comparison with *Transformer Flow Policy*, *EffiSurg* reduces inference latency by 67% while improving the average task success rate from 80.0% to 96.3%.

This efficiency gain is consistent with the decoupled design: action-specific conditioning is extracted once per chunk, allowing subsequent vector-field evaluations to use a lightweight flow expert without repeated interaction with the full multimodal representations. The reduction is achieved with the same number of flow-integration steps as the coupled baseline. The following attention analysis and ablations further examine the architectural design.

C. Visual Attention Analysis

To examine the spatial distribution of attention when extracting action-relevant information, Fig. 7 compares *EffiSurg* and *Transformer Flow Policy* on the same multimodal observations from needle handoff and knot tying. Attention weights are extracted from the final cross-attention layer of the conditioning module in *EffiSurg* and the action expert in the coupled baseline, using action queries and evolving action tokens as queries, respectively. The weights are averaged across all attention heads and action positions, and additionally across all flow-integration steps for the coupled baseline. The entries corresponding to top-view tokens within the multimodal representations are then selected and independently min-max normalized, mapped to their corresponding image-patch locations, and overlaid on the top-view images using a common color scale.

In the illustrated examples, *EffiSurg* concentrates its high-attention regions around the instrument tips and nearby needle or suture interaction sites. In contrast, *Transformer Flow Policy* exhibits a more dispersed distribution, with prominent responses in tissue regions away from the immediate manipulation site. These observations suggest that decoupled multimodal grounding provides more focused visual guidance for action generation, offering a possible explanation for the improved task performance.

TABLE II
EFFISURG ABLATIONS IN PHANTOM AND EX VIVO SETTINGS.

Method	Avg. Success (%) $\uparrow$	Latency (ms) $\downarrow$
<i>EffiSurg</i> w/ Shared Conditioning	82.5	46
<i>EffiSurg</i> w/o Self-Attention	72.5	45
<i>EffiSurg</i>	96.3	49

D. Ablation Studies

Architectural ablations examine action-specific conditioning and intermediate self-attention across the same four needle-handoff and knot-tying tasks in phantom and ex vivo settings as the main experiments. Table II reports the arithmetic mean of success rates across these four tasks and the end-to-end inference latency per action chunk.

Effect of Action-Specific Conditioning. To evaluate action-specific conditioning, *EffiSurg* w/ *Shared Conditioning* replaces the H action queries with a single learned query that extracts one shared conditioning feature from the multimodal representations. This feature is broadcast to all H future-action positions, with a learned, observation-independent positional embedding added at each position before flow refinement. The remaining architecture and training and evaluation settings are unchanged. Replacing action-specific conditioning with this shared alternative decreases the average success rate across the four tasks from 96.3% to 82.5%, while reducing inference latency from 49 to 46 ms. These results support the use of action-specific conditioning to improve overall task performance at a modest additional inference cost.

Effect of Intermediate Self-Attention. The intermediate self-attention layer is removed to evaluate the importance of information exchange among future actions during flow refinement. Without self-attention, each future action is processed primarily through the shared MLP blocks under its corresponding conditioning feature, without explicit interaction with other actions in the predicted chunk. Removing self-attention decreases the average success rate across the four tasks from 96.3% to 72.5%, while reducing inference latency from 49 to 45 ms. These results support the role of intermediate self-attention in modeling cross-action dependencies, with a modest latency overhead and without reintroducing repeated multimodal interaction.

Effect of Flow Integration Steps. Finally, the influence of the number of flow-integration steps is investigated by varying K while keeping the remaining model configuration unchanged. Fig. 8 reports both average task success rate and inference latency under different values of K . Increasing the number of integration steps initially improves average task success rate but also increases inference latency. Performance saturates around $K = 10$, providing a favorable balance between manipulation performance and computational efficiency.

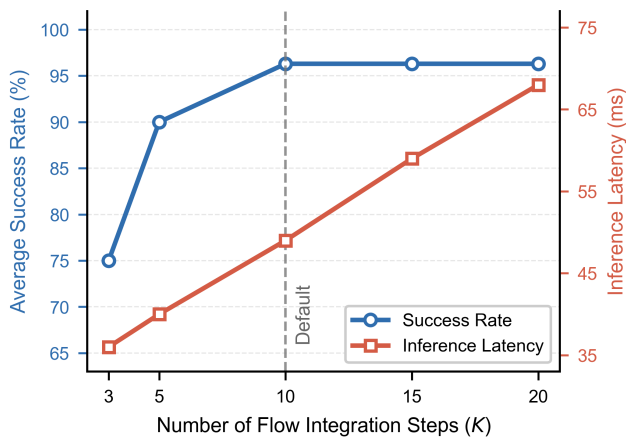


Fig. 8. Effect of the number of flow-integration steps K on average success rate and end-to-end inference latency. Success rate saturates around $K = 10$, while latency continues to increase; the dashed line marks the default setting $K = 10$.

VI. CONCLUSION

This paper presented EffiSurg, an efficient flow-matching policy for autonomous microsurgical manipulation. By extracting action-specific conditioning features once per action chunk and reusing them in a lightweight flow expert, EffiSurg decouples multimodal grounding from iterative action refinement while preserving multi-step generation. Across four needle-handoff and knot-tying tasks in phantom and ex vivo settings, EffiSurg achieved an average success rate of 96.3% and an end-to-end policy inference latency of 49 ms per action chunk. Compared with the coupled Transformer Flow Policy baseline, it improved success by 16.3 percentage points and reduced latency by 67% using the same number of flow-integration steps. Architectural ablations further support the contributions of action-specific conditioning and intermediate self-attention to task performance. These phantom and ex vivo experiments evaluate task performance but do not establish clinical safety. Future work will investigate more diverse surgical tasks and robustness under greater environmental variability.

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